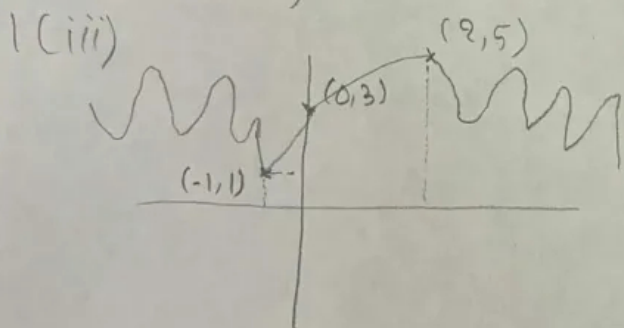


Assignment 4

1(i) a) yes, $f(x) = e^x$ works since $f'(x) = e^x > 0$ and $f''(x) = e^x > 0$ for all x and $\lim_{x \rightarrow -\infty} e^x = 0$ so $y = 0$ is a horizontal asymptote at $-\infty$

b) No, $f''(x) > 0$ for all x , the derivative $f'(x)$ is increasing on $\mathbb{R}$. $f'(x) > 0$ for all x - for every $x > x_0$ number so $f'(x) \geq f'(x_0) > 0$. Mean Value Theorem for $x > x_0$ so $f(x) - f(x_0) > f'(x_0)(x - x_0)$ as $x \rightarrow \infty$ so $f(x) \rightarrow \infty$ so f can't approach a finite horizontal asymptote as $x \rightarrow \infty$

1(ii) an example is $f(x) = \ln(x+1)$, $x > -1$ then $f'(x) = \frac{1}{x+1} > 0$ and $f''(x) = -\frac{1}{(x+1)^2} < 0$ and for all $x > -1$ so $\lim_{x \rightarrow \infty} \ln(x+1) = \infty$



2(i) we have $f(x) = e^{-x} \ln x$
 so derivative is:

$$f'(x) = (-e^{-x}) \ln x + e^{-x} \cdot \frac{1}{x}$$

$$= e^{-x} \left(\frac{1}{x} - \ln x \right)$$

Define $g(x) = \frac{1}{x} - \ln x$ so g is continuous on $[1, 3]$ so
 $g(1) = 1 - \ln 1 = 1 > 0$ and $g(3) = \frac{1}{3} - \ln 3 < 0$ since $\ln 3 > \frac{1}{3}$
 so by Intermediate Value Theorem, g is continuous on $[1, 3]$ and changes sign so $c \in (1, 3)$

where $g(c) = 0$ so $f'(c) = e^{-c} g(c) = 0$ so c is a critical number of f in $(1, 3)$

2) ii) Critical number is a local maximum so since $f'(x) = e^{-x} (\frac{1}{x} - \ln x)$ and $e^{-x} > 0$, sign of $f'(x)$ is same as $g(x) = \frac{1}{x} - \ln x$ so $g'(x) = -\frac{1}{x^2} - \frac{1}{x} < 0$ so g is decreasing. Since $g(1) > 0$ and $g(3) < 0$ so critical number satisfies $g(x) > 0$ for $x < c$, $g(x) < 0$ for $x > c$ so $f'(x) > 0$ for $x < c$, $f'(x) < 0$ for $x > c$ so f increases before c and decreases after c so it's a local maximum by First Derivative Test

3) A) $\sin$ and $\cos$ are real numbers, so domain(f) = $\mathbb{R}$ so $0 \leq x \leq 2\pi$; B) $f(0) = \sin 0 + \cos 0 = 1$ so y intercept is $(0, 1)$

C) $f(-x) = \sin(-2x) + \cos(-2x) = -\sin(2x) + \cos(2x)$ f is not odd or not even, it's periodic π .

D) No vertical asymptotes because $\sin$ and $\cos$ defined in $\mathbb{R}$ everywhere and no horizontal asymptotes because it's periodic and oscillates. E) we have:

$$f'(x) = 2\cos(2x) - 2\sin(2x) = 2(\cos(2x) - \sin(2x))$$

and critical points are at $\cos(2x) - \sin(2x) = 0$

$$\text{so } 2x = \frac{\pi}{4} + k\pi, \quad x = \frac{\pi}{8} + \frac{k\pi}{2}$$

inflection points at $(\frac{3\pi}{8}, 0)$, $(\frac{7\pi}{8}, 0)$, $(\frac{11\pi}{8}, 0)$, $(\frac{15\pi}{8}, 0)$

2) ii) So

$g'(x)$

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